Assignment 1

CS 6117 (Fall 2025)

Exercise 1. Let $\mathbf{C}$ and $\mathbf{D}$ be categories. The product category $\mathbf{C} \times \mathbf{D}$ has as objects pairs of objects from $\mathbf{C}$ and $\mathbf{D}$.

Unpack what this definition means, i.e., spell out the definition of $\mathbf{C} \times \mathbf{D}$ and show that it is a category.

The product of $\mathbf{C}$ and $\mathbf{D}$ has objects pairs (C, D) where C is an object of $\mathbf{C}$ and D an object of $\mathbf{D}$. The morphisms $(C, D) \to (C', D')$ are pairs of morphisms (f, g) where $f : C \to C'$ and $g : D \to D'$ with composition given coordinatewise. For each object (C, D), the identity morphism is $(\mathrm{id}_C, \mathrm{id}_D)$, where id_C and id_D are the identities in $\mathbf{C}$ and $\mathbf{D}$. Since composition is associative in $\mathbf{C}$ and $\mathbf{D}$, for any morphism $(f, g) : (C_1, D_1) \to (C_2, D_2)$,$(f, g) \circ (\mathrm{id}_{C_1}, \mathrm{id}_{D_1}) = (f \circ \mathrm{id}_{C_1}, g \circ \mathrm{id}_{D_1}) = (f, g)$, and similarly for other identity. Thus, $\mathbf{C} \times \mathbf{D}$ is a category.

Exercise 2. Let **Met** be the category of *metric spaces*, where objects (X, d_X) are sets X equipped with a distance function $d_X : X \times X \to \mathbb{R}$ satisfying the metric axioms, and morphisms from (X, d_X) to (Y, d_Y) are *non-expansive* maps: functions $f : X \to Y$ such that $d_Y(f(x_1), f(x_2)) \leq d_X(x_1, x_2)$.

1. Given metric spaces (X, d_X) and (Y, d_Y), show that the metric space with carrier set $X \times Y$ and the max distance:
$$d_{max}((x_1, y_1), (x_2, y_2)) \triangleq \max(d_X(x_1, x_2), d_Y(y_1, y_2))$$
is a Cartesian product for (X, d_X) and (Y, d_Y).

   Think that for any metric space (Z, d_Z) and non-expansive maps $f : Z \to X$, $g : Z \to Y$, there exists a unique non-expansive map $h : Z \to X \times Y$ such that $\pi_X \circ h = f$ and $\pi_Y \circ h = g$. h is determined by its morphisms. $h(z) = (f(z), g(z))$ satisfies the commuting conditions. It is non-expansive because

   $$d_{\max}(h(z_1), h(z_2)) = \max(d_X(f(z_1), f(z_2)), d_Y(g(z_1), g(z_2))) \leq \max(d_Z(z_1, z_2), d_Z(z_1, z_2)) = d_Z(z_1, z_2)$$

   since f and g are non-expansive. $d_X(\pi_X(p_1), \pi_X(p_2)) = d_X(x_1, x_2) \leq d_{\max}(p_1, p_2)$ and similarly for π_Y. Showing that $(X \times Y, d_{\max})$ with $\pi_X : (x, y) \mapsto x$ and $\pi_Y : (x, y) \mapsto y$ is a Cartesian product in **Met**.

2. Given metric spaces (X, d_X) and (Y, d_Y), show that the metric space with carrier set $X \times Y$ and the sum distance:
$$d_{sum}((x_1, y_1), (x_2, y_2)) \triangleq d_X(x_1, x_2) + d_Y(y_1, y_2)$$
along with the usual first and second projection morphisms is usually *not* a Cartesian product for (X, d_X) and (Y, d_Y) (outside of a few special cases, like when X and Y are empty or singleton).

   Considering $X = Y = \mathbb{R}$ with standard metric $d(a, b) = |a - b|$. Suppose $(X \times Y, d_{sum})$ were a product. Then, for $Z = \mathbb{R}$ with standard metric, and maps $f(z) = z$, $g(z) = z$ (both non-expansive), then $h(z) = (z, z)$ should be non-expansive $d_{sum}(h(z_1), h(z_2)) = |z_1 - z_2| + |z_1 - z_2| = 2|z_1 - z_2| \not\leq |z_1 - z_2|$ unless $|z_1 - z_2| = 0$. Thus, h is not non-expansive, so $(X \times Y, d_{sum})$ is not a product.

Exercise 3. Let **C** be a category with products. Then, $A \times B$ is naturally isomorphic to $B \times A$. Unpack what this statement means, i.e., define all categories and functors that are involved, and then prove that these functors are naturally isomorphic.

Note that it is *not* enough to show that $A \times B$ is isomorphic to $B \times A$ for all objects A and B: the isomorphism should be natural in the technical sense.

Let $\mathcal{C}$ be a category with products, meaning for any objects $A, B \in \mathcal{C}$, there exists a product object $A \times B$ with projection morphisms $\pi_1 : A \times B \to A$ and $\pi_2 : A \times B \to B$, satisfying the universal property. Define the functor $A \times - : \mathcal{C} \to \mathcal{C}$ taking B to $A \times B$ and morphism $f : B \to C$ to $A \times f : A \times B \to A \times C$. Similarly, $- \times A : \mathcal{C} \to \mathcal{C}$ and $f \times A : B \times A \to C \times A$. Define $\tau_B : A \times B \to B \times A$ as the unique morphism induced by mapping $\pi_1 : A \times B \to A$ to $\pi_2 : B \times A \to A$ and $\pi_2 : A \times B \to B$ to $\pi_1 : B \times A \to B$. The inverse $\sigma_B : B \times A \to A \times B$ is defined analogously. The compositions $\sigma_B \circ \tau_B$ and $\tau_B \circ \sigma_B$ are identities, confirming τ_B is an isomorphism. Naturality holds as $\tau_C \circ (A \times f) = (f \times A) \circ \tau_B$, verified by the universal property of products.

$$\begin{array}{ccc} A \times B & \xrightarrow{\tau_B} & B \times A \\ {\scriptstyle A \times f}\downarrow & & \downarrow{\scriptstyle f \times A} \\ A \times C & \xrightarrow{\tau_C} & C \times A \end{array}$$

Exercise 4. A real vector space is a set V with a binary addition operation $+ : V \times V \to V$, an identity element $0 \in V$, and a unary scalar operation $r \cdot (-) : V \to V$ for every $r \in \mathbb{R}$ satisfying some equations. When dealing with finite-dimensional vector spaces, it is useful to think of them as tuples of real numbers. In this exercise we will show that there is a category-theoretical justification for this kind of reasoning.

Let **FDVect** be the category of finite-dimensional real vector spaces and linear transformations, and let **Mat** be the category where objects are $n \in \mathbb{N}$ and morphisms $M : n \to m$ are $n \times m$ matrices with real entries. For the following exercises you may freely use basic facts from linear algebra.

1. Define the identity morphism and composition in **Mat**, and show that your definitions make **Mat** into a category.

   For each $n \in \mathbb{N}$, the identity morphism $id_n : n \to n$ is the $n \times n$ identity matrix. Given morphisms $M : n \to m$ (an $m \times n$ matrix) and $N : m \to p$ (a $p \times m$ matrix), their composition $N \circ M : n \to p$ is the $p \times n$ matrix product NM. Matrix multiplication is associative. For any $M : n \to m$, $MI_n = M$ and $I_m M = M$, by properties of matrix multiplication. Thus, **Mat** is a category.

2. Show that **FDVect** and **Mat** are equivalent categories.

   Define functors $F : \mathbf{FDVect} \to \mathbf{Mat}$ and $G : \mathbf{Mat} \to \mathbf{FDVect}$ forming a natural equivalence, $\eta : id_{\mathbf{FDVect}} \to GF$ and $\epsilon : FG \to id_{\mathbf{Mat}}$ are natural isomorphisms. For a linear map $f : V \to W$, choose bases $\mathcal{B}_V$ for V and $\mathcal{B}_W$ for W; then $F(f)$ is the $[\dim W \times \dim V]$-matrix of f with respect to these bases. For $V \in \mathbf{FDVect}$, let $\eta_V : V \to GF(V) = \mathbb{R}^{\dim V}$ be the isomorphism given by coordinates with respect to a basis $\mathcal{B}_V$ of V. For $M : n \to m$, $G(M)(v) = Mv$. For $n \in \mathbf{Mat}$, $\epsilon_n : FG(n) = \dim(\mathbb{R}^n) = n \to n$ is the identity matrix I_n. These preserve identities and composition by linear algebra: $F(id_V) = I_{\dim V}$, $G(I_n) = id_{\mathbb{R}^n}$, and $F(g \circ f) = F(g)F(f)$, $G(N \circ M) = G(N) \circ G(M)$. Since η and ϵ are natural isomorphisms (η_V is basis-isomorphism, $\epsilon_n = id_n$), $F \dashv G$ is an equivalence of categories. Thus, $\mathbf{FDVect} \simeq \mathbf{Mat}$.

3. Show that **FDVect** and **Mat** are not isomorphic categories: there are not functors $F : \mathbf{FDVect} \to \mathbf{Mat}$ and $G : \mathbf{Mat} \to \mathbf{FDVect}$ such that $F \circ G = Id_{\mathbf{Mat}}$ and $G \circ F = Id_{\mathbf{FDVect}}$.

   Mat has objects natural numbers, countable, while **FDVect** has uncountably many objects, so no bijection on objects, hence no isomorphism of categories.

Exercise 5. Let $\mathbf{C}$ be a category. The *opposite category* $\mathbf{C}^{op}$ has the same objects as $\mathbf{C}$, but morphisms in the opposite direction: $\mathrm{Hom}_{\mathbf{C}^{op}}(A, B) \triangleq \mathrm{Hom}_{\mathbf{C}}(B, A)$.

1. Unpack the definition of $\mathbf{C}^{op}$. How is composition defined?

   The opposite category $\mathbf{C}^{op}$ has the same objects as $\mathbf{C}$, and the collections of morphisms of $\mathbf{C}^{op}$ is the same as the collection of morphisms of $\mathbf{C}$ but reversed. Composition in $\mathbf{C}^{op}$ is defined as given morphisms $f : A \to B$ (which corresponds to $f : B \to A$ in $\mathbf{C}$) and $g : B \to C$ (corresponding to $g : C \to B$ in $\mathbf{C}$), the composition $f \circ g : C \to A$.

2. We saw the image/pushforward functor $P^* : \mathbf{Set} \to \mathbf{Set}$ in class. There is also an inverse image/pullback functor $P_* : \mathbf{Set}^{op} \to \mathbf{Set}$. Define this functor. (You don't have to spell out all the properties but you should check them for yourself.)

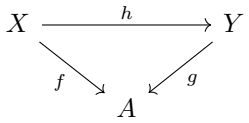
   Consider a morphism $f^{\mathrm{op}} : B \to A$ in $\mathbf{Set}^{op}$, which corresponds to a function $f : A \to B$ in $\mathbf{Set}$. Define $P(f^{op}) : \mathcal{P}(B) \to \mathcal{P}(A)$ by $P(f^{\mathrm{op}})(B') = f^{-1}(B') = \{a \in A \mid f(a) \in B'\}$ for any subset $B' \subseteq B$. For the identity morphism $id_A^{\mathrm{op}} : A \to A$ in $\mathbf{Set}^{\mathrm{op}}$ (corresponding to $id_A : A \to A$), $P(id_A^{\mathrm{op}})(S) = id_A^{-1}(S) = S$ for any $S \subseteq A$, so it is the identity on $\mathcal{P}(A)$. Let $f^{\mathrm{op}} : B \to A$ and $g^{\mathrm{op}} : C \to B$ in $\mathbf{Set}^{\mathrm{op}}$ (corresponding to $f : A \to B$, $g : B \to C$ in $\mathbf{Set}$). The composition is $f^{\mathrm{op}} \circ g^{\mathrm{op}} = (g \circ f)^{\mathrm{op}} : C \to A$. Then $P(f^{\mathrm{op}} \circ g^{\mathrm{op}}) = (g \circ f)^{-1} : \mathcal{P}(C) \to \mathcal{P}(A)$. $P(f^{\mathrm{op}}) \circ P(g^{\mathrm{op}}) = f^{-1} \circ g^{-1} : \mathcal{P}(C) \to \mathcal{P}(B) \to \mathcal{P}(A)$. For any $U \subseteq C$, $f^{-1}(g^{-1}(U)) = \{a \in A \mid f(a) \in g^{-1}(U)\} = \{a \in A \mid g(f(a)) \in U\} = (g \circ f)^{-1}(U)$, so functor $P_* : \mathbf{Set}^{op} \to \mathbf{Set}$ is defined.

Functors from $\mathbf{C}^{op}$ to $\mathbf{C}$ are sometimes called *contravariant*, because they reverse the direction of morphisms. Another example is the *dagger* functor $(-)^\dagger : \mathbf{Rel}^{op} \to \mathbf{Rel}$, which is identity on objects and maps relations to relation inverse.

Exercise 6 (Category of the Week). Let $\mathbf{C}$ be a category and A be any object in $\mathbf{C}$. The *slice category* $\mathbf{C}/A$, also known as the *over* category, has as objects morphisms $X \xrightarrow{f} A$ in $\mathbf{C}$, and morphisms being "commuting triangles".

1. Unpack what this means: spell out the definition of $\mathbf{C}/A$ and show that it forms a category.

   Let $\mathcal{C}$ be a category and A an object in $\mathcal{C}$. The slice category $\mathcal{C}/A$ is defined as follows: Morphism $f : X \to A$ where X is an object in $\mathcal{C}$. Given objects $(X, f : X \to A)$ and $(Y, g : Y \to A)$, a morphism $h : (X, f) \to (Y, g)$ is a morphism $h : X \to Y$ in $\mathcal{C}$ such that the triangle commutes: $g \circ h = f$,

   For morphisms $h : (X, f) \to (Y, g)$ and $k : (Y, g) \to (Z, l)$, their composition $k \circ h : (X, f) \to (Z, l)$ is the composition $k \circ h$ in $\mathcal{C}$, which commutes because $l \circ (k \circ h) = (l \circ k) \circ h = g \circ h = f$. For an object (X, f), the identity morphism is $id_X : X \to X$ in $\mathcal{C}$, since $f \circ id_X = f$. Confirming $\mathbf{C}/A$ is a category.

2. Let A, B be objects in $\mathbf{C}$. A morphism $f : A \to B$ "induces" a functor $f : \mathbf{C}/A \to \mathbf{C}/B$. Unpack what this means: spell out the definition and show that f is a functor.

   Let A, B be objects in $\mathcal{C}$, and $f : A \to B$ a morphism. For an object $(X, g : X \to A)$ in $\mathcal{C}/A$, $f_*(X, g) = (X, f \circ g : X \to B)$ in $\mathcal{C}/B$. For a morphism $h : (X, g) \to (Y, k)$ in $\mathcal{C}/A$ (so $h : X \to Y$, $k \circ h = g$), $f_*(h) = h : (X, f \circ g) \to (Y, f \circ k)$, since $(f \circ k) \circ h = f \circ (k \circ h) = f \circ g$. $f_*(id_{(X,g)}) = id_X = id_{f_*(X,g)}$. For $p \circ q$ in $\mathcal{C}/A$, $f_*(p \circ q) = p \circ q = f_*(p) \circ f_*(q)$ in $\mathcal{C}/B$. This induces a functor $f_* : \mathcal{C}/A \to \mathcal{C}/B$.

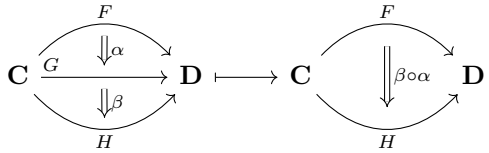
3. Let A be a set. Show that $\mathbf{Set}/A$ has products and coproducts. (We won't get to limits and colimits for a while, but $\mathbf{C}/A$ has limits/colimits whenever $\mathbf{C}$ has limits/colimits, respectively.)

   Let A be a set. The category $\mathbf{Set}/A$ has objects $(X, f : X \to A)$ (functions to A), and morphisms $h : (X, f) \to (Y, g)$ with $g \circ h = f$. For products in $\mathbf{Set}/A$: Given objects $(X, f), (Y, g)$ in $\mathbf{Set}/A$, their product is (Z, p) where $Z = \{(x, y) \in X \times Y \mid f(x) = g(y)\}$, and $p : Z \to A$ by $p(x, y) = f(x)$ (or equivalently $g(y)$). $\pi_X : Z \to X$, $(x, y) \mapsto x$; $\pi_Y : Z \to Y$, $(x, y) \mapsto y$. These are morphisms in $\mathbf{Set}/A$ because $f \circ \pi_X = p = g \circ \pi_Y$. For any $(W, q : W \to A)$ with morphisms $u : (W, q) \to (X, f)$, $v : (W, q) \to (Y, g)$ (so $f \circ u = q = g \circ v$), the unique map $m : W \to Z$, $w \mapsto (u(w), v(w))$, is well-defined because $f(u(w)) = q(w) = g(v(w))$, and it is a morphism since $p \circ m = q$.

   For coproducts in $\mathbf{Set}/A$: The coproduct of $(X, f), (Y, g)$ is $(X + Y, [f, g])$ and $[f, g] : X + Y \to A$ is the unique morphism such that $[f, g] \circ \iota_X = f$, $[f, g] \circ \iota_Y = g$. $i_X = \iota_X : (X, f) \to (X + Y, [f, g])$, $i_Y = \iota_Y : (Y, g) \to (X + Y, [f, g])$, which are morphisms since the compositions match by definition. For any (W, q) with morphisms $u : (X, f) \to (W, q)$, $v : (Y, g) \to (W, q)$ (so $q \circ u = f$, $q \circ v = g$), the unique $m : X + Y \to W$ such that $m \circ \iota_X = u$, $m \circ \iota_Y = v$ from the coproduct universal property in $\mathbf{Set}$, and it is a morphism in $\mathbf{Set}/A$ because $q \circ m = [q \circ u, q \circ v] = [f, g]$.

Optional Exercises. For some more practice calculations, the following exercises work out some standard ways to compose natural transformations.

1. Let $F, G, H : \mathbf{C} \to \mathbf{D}$ be functors, and let $\alpha : F \Rightarrow G$ and $\beta : G \Rightarrow H$ be natural transformations. The following diagram depicts the $\beta \circ \alpha$, the *vertical composition* of α and β:

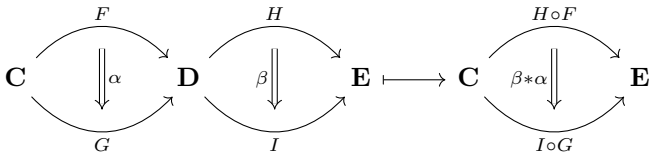


   Spell out the definition of $\beta \circ \alpha$ and show that it is indeed a natural transformation.

   Let $F, G, H : \mathbf{C} \to \mathbf{D}$ be functors, and let $\alpha : F \to G$ and $\beta : G \to H$ be natural transformations. The vertical composition $\beta \circ \alpha$ is defined componentwise as $(\beta \circ \alpha)_C = \beta_{G(C)} \circ \alpha_C : F(C) \to H(C)$ for each $C \in \mathbf{C}$.

   Considering any morphism $f : C \to C'$ in $\mathbf{C}$. To show the commutativity of the diagram $\begin{array}{ccc} F(C) & \xrightarrow{(\beta\circ\alpha)_C} & H(C) \\ \downarrow{\scriptstyle F(f)} & & \downarrow{\scriptstyle H(f)} \\ F(C') & \xrightarrow{(\beta\circ\alpha)_{C'}} & H(C') \end{array}$ where $(\beta \circ \alpha)_C = \beta_{G(C)} \circ \alpha_C$ and $(\beta \circ \alpha)_{C'} = \beta_{G(C')} \circ \alpha_{C'}$. The left path is $(\beta \circ \alpha)_{C'} \circ F(f)$, and the right path is $H(f) \circ (\beta \circ \alpha)_C$. Using the naturality of α, the diagram $\begin{array}{ccc} F(C) & \xrightarrow{\alpha_C} & G(C) \\ \downarrow{\scriptstyle F(f)} & & \downarrow{\scriptstyle G(f)} \\ F(C') & \xrightarrow{\alpha_{C'}} & G(C') \end{array}$ commutes, so $\alpha_{C'} \circ F(f) = G(f) \circ \alpha_C$. Applying the naturality of β, $\begin{array}{ccc} G(C) & \xrightarrow{\beta_{G(C)}} & H(C) \\ \downarrow{\scriptstyle G(f)} & & \downarrow{\scriptstyle H(f)} \\ G(C') & \xrightarrow{\beta_{G(C')}} & H(C') \end{array}$ commutes, so $\beta_{G(C')} \circ G(f) = H(f) \circ \beta_{G(C)}$. Composing, $\beta_{G(C')} \circ (\alpha_{C'} \circ F(f)) = \beta_{G(C')} \circ (G(f) \circ \alpha_C)$. The left side is $(\beta_{G(C')} \circ \alpha_{C'}) \circ F(f) = (\beta \circ \alpha)_{C'} \circ F(f)$, and the right side is $(\beta_{G(C')} \circ G(f)) \circ \alpha_C = (H(f) \circ \beta_{G(C)}) \circ \alpha_C = H(f) \circ (\beta_{G(C)} \circ \alpha_C) = H(f) \circ (\beta \circ \alpha)_C$. Thus, the diagram commutes, proving $\beta \circ \alpha$ is natural.

2. Let $F, G : \mathbf{C} \to \mathbf{D}$ and $H, I : \mathbf{D} \to \mathbf{E}$ be functors, and let $\alpha : F \Rightarrow G$ and $\beta : H \Rightarrow I$ be natural transformations. The following diagram depicts the $\beta * \alpha$, the *horizontal composition* of α and β:

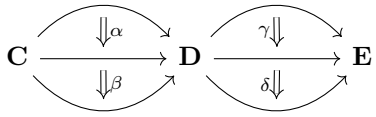


Spell out the definition of $\beta * \alpha$ and show that it is indeed a natural transformation.

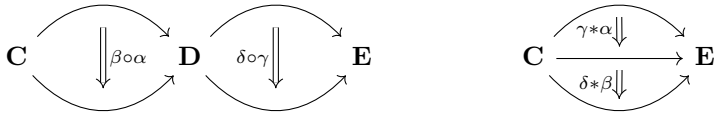
Let $F, G : \mathbf{C} \to \mathbf{D}$ and $H, I : \mathcal{D} \to \mathbf{E}$ be functors, and let $\alpha : F \to G$ and $\beta : H \to I$ be natural transformations. The horizontal composition $\beta * \alpha$ is a natural transformation from $H \circ F$ to $I \circ G$, with component $(\beta * \alpha)_C = \beta_{G(C)} \circ H(\alpha_C) : (H \circ F)(C) \to (I \circ G)(C)$ for each $C \in \mathbf{C}$.

Considering $f : C \to C'$ in $\mathcal{C}$. The diagram $\begin{array}{ccc} (H \circ F)(C) & \xrightarrow{(\beta * \alpha)_C} & (I \circ G)(C) \\ \downarrow{\scriptstyle (H \circ F)(f)} & & \downarrow{\scriptstyle (I \circ G)(f)} \\ (H \circ F)(C') & \xrightarrow{(\beta * \alpha)_{C'}} & (I \circ G)(C') \end{array}$ must commute. The left path is $(\beta * \alpha)_{C'} \circ (H \circ F)(f) = \beta_{G(C')} \circ H(\alpha_{C'}) \circ H(F(f))$, and the right path is $(I \circ G)(f) \circ (\beta * \alpha)_C = I(G(f)) \circ \beta_{G(C)} \circ H(\alpha_C)$. Using α's naturality, $G(f) \circ \alpha_C = \alpha_{C'} \circ F(f)$, so $H(G(f)) \circ H(\alpha_C) = H(\alpha_{C'}) \circ H(F(f))$. With β's naturality, $I(G(f)) \circ \beta_{G(C)} = \beta_{G(C')} \circ H(G(f))$, $\beta_{G(C')} \circ H(\alpha_{C'}) \circ H(F(f)) = I(G(f)) \circ \beta_{G(C)} \circ H(\alpha_C)$. Thus, $\beta * \alpha$ is natural.

3. Natural transformations satisfy the *interchange law*. Given functors and natural transformations as follows:



show that $(\delta \circ \gamma) * (\beta \circ \alpha) = (\delta * \beta) \circ (\gamma * \alpha)$. In pictures, the following two natural transformations are equal:



Given functors $F, G, H, I, J : \mathcal{C} \to \mathcal{D}$ and natural transformations $\alpha : F \to G$, $\beta : G \to H$, $\gamma : H \to I$, $\delta : I \to J$, then $(\delta \circ \gamma) * (\beta \circ \alpha) = (\delta * \beta) \circ (\gamma * \alpha)$. Define $(\delta \circ \gamma)_C = \delta_{I(C)} \circ \gamma_C$, $(\beta \circ \alpha)_C = \beta_{G(C)} \circ \alpha_C$. Then $[(\delta \circ \gamma) * (\beta \circ \alpha)]_C = (\delta \circ \gamma)_{H(C)} \circ J(\beta_{G(C)} \circ \alpha_C) = \delta_{I(H(C))} \circ \gamma_{H(C)} \circ J(\beta_{G(C)}) \circ J(\alpha_C)$. For the right side, $(\delta * \beta)_C = \delta_{G(C)} \circ I(\beta_C)$, $(\gamma * \alpha)_C = \gamma_{G(C)} \circ J(\alpha_C)$, so $[(\delta * \beta) \circ (\gamma * \alpha)]_C = (\delta_{G(G(C))} \circ I(\beta_{G(C)})) \circ (\gamma_{G(C)} \circ J(\alpha_C)) = \delta_{G(G(C))} \circ I(\beta_{G(C)}) \circ \gamma_{G(C)} \circ J(\alpha_C)$. Using naturality of β, $H(f) \circ \beta_C = \beta_{C'} \circ G(f)$, and γ, $I(f) \circ \gamma_C = \gamma_{C'} \circ H(f)$, the compositions align. The interchange law holds as naturality squares commute, confirming equality.

These diagrams are also known as *pasting diagrams*. The interchange law, combined with a few other laws satisfied by natural transformations, ensures that any well-formed pasting diagram is *unambiguous*: the various ways of converting a diagram into horizontal and vertical compositions all encode the same natural transformation.